

Announcements

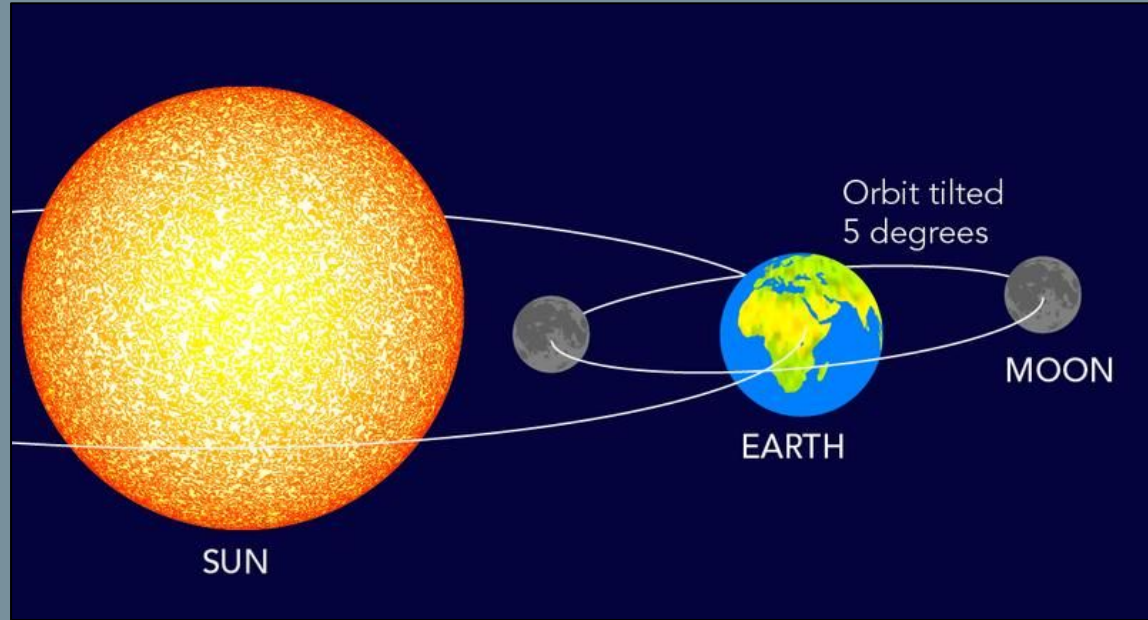
- Extra Credit Opportunity Tuesday 9/1 and Thursday 9/3, Both at 7:30 P.M.
 - “Coordinate Systems” show in Marston Theater in ISTB4
 - It will give additional information for next week’s *Coordinate Systems Lab*
 - There will be sign in sheets to track attendance for extra credit purposes
 - You just need to show up and watch the show
- Drop/Add dates have passed - if any new students are here, please talk to me after class to get caught up

Lunar Phases

(and solar eclipses)

The Sun-Earth-Moon System

- The Earth orbits the Sun, and the Moon orbits the Earth
- The Moon's orbit is "tilted" about 5° relative to the Earth's orbital plane around the Sun
- The Moon is tidally locked to the Earth
 - Meaning it rotates once for every full orbit it makes
 - (*hint* Remember this *hint*)



Not to scale

Phases

- As you likely know, the Moon has different phases
- These are caused by the relative positions of the Moon and Sun
- The Moon is always half illuminated (not counting lunar eclipses)
- But we can't always see all of that half from Earth

Moon Phases

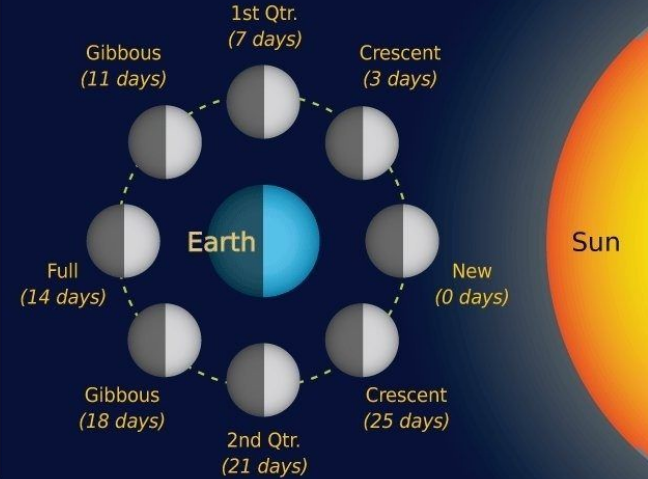
(Not to scale)

As Viewed from Earth



Waxing → getting bigger
Waning → getting smaller

Moon's orbit around Earth



Venus and Mercury also have phases! →
This is true of anything that moves between the Earth and Sun



Tonight's Lab (Part 1)

- Break into groups, assigning one person the role of “Sun”, another the role of “Earth”, and a third the role of “Moon”
 - The Sun will need a flashlight - your cellphone's flashlight should be fine
 - The Earth will need two pieces of paper - one labeled “West” and another labeled “East”
 - The Moon will need a pen or pencil to act as a handle for the Moon model
- First, the Earth and Sun will explore the relationship between the Earth's rotation and the time of day
- Then, you will include the Moon and see how many of its “sides” are illuminated by the Sun, and how many of those are visible from Earth
 - You will do this several times with different rotation rates of the Moon
- This part goes through question 20

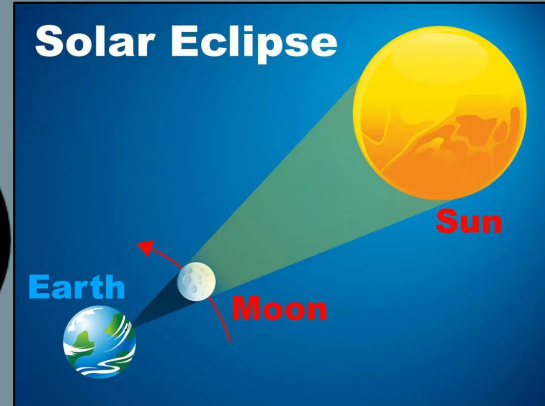
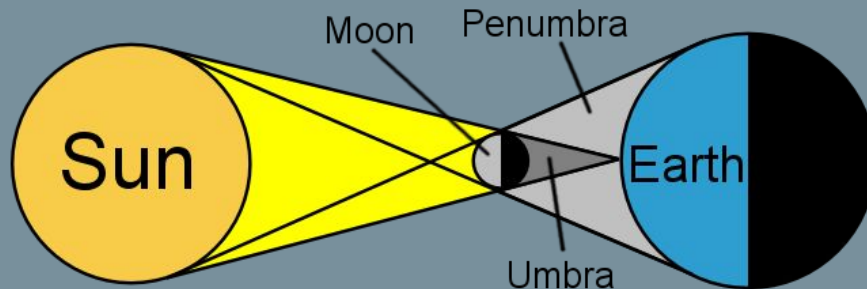
Tonight's Lab (Part 2)

- You are going to calculate how much of each side is illuminated during each phase
- Then graph how the % (of each hemisphere) illuminated changes over time
- This part goes through question 25

$$\text{Illumination (\%)} = \frac{\text{Segments illuminated}}{\text{Total Hemisphere Segments}} \times 100$$

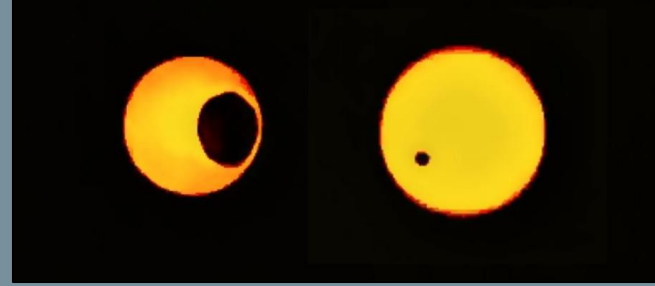
Solar Eclipses

- There is a possibility of a solar eclipse occurring during the observing window for the VR experience's lunar telescope
- But we don't see it happen every month - because of the Moon's 5° orbital tilt
 - Instead, it lines up about every 6 months
- A solar eclipse occurs when the Moon passes in between the Earth and Sun, blocking some or all of the Sun's light - meaning it can only happen during a **New Moon** (Lunar eclipses happen during a Full Moon)

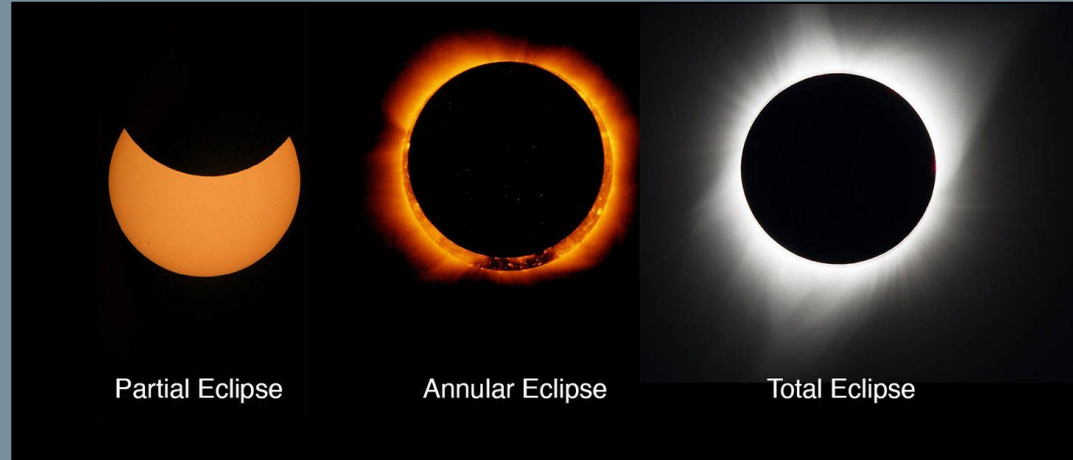


Solar Eclipses and Angular Size

- There are different types of solar eclipse
 - Partial - Only a part of the Sun is covered
 - Annular - The Sun would be entirely covered, but the Moon isn't quite big enough
 - Total - The Moon fully covers the Sun
- The fact that the Moon can cover the Sun, is because they have (roughly) the same angular size - despite dramatically different actual sizes
- Moon's orbit is not perfectly circular, so, it is not always the same distance, and therefore not always the same angular size
 - Hence sometimes annular and sometimes total eclipses



↑ Sad Martian eclipses ↑



Tonight's Lab (Part 3)

- Answer questions 26-28
- Calculate the angular sizes of the Sun and Moon
- Use your results from the earlier parts to determine if the Solar Eclipse could impact the telescope's observations
- This will take through the end of the lab (question 38)

$$\theta = 2.06 \cdot 10^5 \cdot \frac{D}{d},$$

Table 2: Distances and diameters of the Sun and the Moon during the Lunar Telescope observation window.

Object	Distance from Earth	Diameter
Sun	1.50×10^8 km	1.39×10^6 km
Moon	3.68×10^5 km	3.47×10^3 km

Image Credits (In order of appearance)

- Let's Talk Science
 - <https://letstalkscience.ca/educational-resources/backgrounders/earth-moon-system>
- University of Iowa Pressbooks
 - <https://pressbooks.uiowa.edu/science-methods-iii/chapter/earth/>
- BBC Sky at Night Magazine
 - <https://www.skyatnightmagazine.com/advice/venus-dichotomy>
- NASA
 - <https://www.nasa.gov/image-article/solar-eclipse-diagram/>
- Khan Academy
 - <https://www.khanacademy.org/science/cosmology-and-astronomy/earth-history-topic/moon-phases-and-eclipses/a/solar-eclipse-2017>
- NASA Jet Propulsion Laboratory
 - <https://www.jpl.nasa.gov/edu/resources/teachable-moment/the-science-of-solar-eclipses-and-how-to-watch-with-nasa/>
- Forbes
 - <https://www.forbes.com/sites/startswithabang/2020/06/29/nasa-rover-captures-what-a-solar-eclipse-looks-like-on-mars/>